

Polynomials, factor and remainder theorems - Set 1 - Solutions

Specification MM1.6.

Answer key

Question	Answer	Topic
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Worked solutions

Question 1 POL-01 · Polynomials, factor and remainder theorems · spec MM1.6

When $x^3 - 4x^2 + kx + 6$ is divided by $(x - 3)$ the remainder is -6 . Find k .

Answer: F

Remainder Theorem: $27 - 36 + 3k + 6 = -6 \Rightarrow 3k - 3 = -6 \Rightarrow k = -1$.

Question 2 POL-02 · Polynomials, factor and remainder theorems · spec MM1.6

$(x + 2)$ is a factor of $2x^3 + ax^2 - 5x - 6$. Find the value of a .

Answer: F

Factor Theorem: $f(-2) = -16 + 4a + 10 - 6 = 4a - 12 = 0$, so $a = 3$.

Question 3 POL-03 · Polynomials, factor and remainder theorems · spec MM1.6

Fully factorise $x^3 - 7x + 6$.

Answer: B

$f(1) = 0$, so $(x - 1)$ is a factor: $x^3 - 7x + 6 = (x - 1)(x^2 + x - 6) = (x - 1)(x + 3)(x - 2)$.

Question 4 POL-04 · Polynomials, factor and remainder theorems · spec MM1.6

Find the remainder when $x^4 + 2x^3 - 3x + 1$ is divided by $(2x - 1)$.

Answer: F

$f(\frac{1}{2}) = \frac{1}{16} + \frac{2}{8} - \frac{3}{2} + 1 = \frac{1}{16} + \frac{4}{16} - \frac{24}{16} + \frac{16}{16} = -\frac{3}{16}$.

Question 5 POL-05 · Polynomials, factor and remainder theorems · spec MM1.6

The polynomial $p(x) = x^3 + ax^2 + bx + 4$ has $(x - 1)$ and $(x + 2)$ as factors. Find the third linear factor.

Answer: B

Write $p(x) = (x - 1)(x + 2)(x - r)$. The constant term is $(-1)(2)(-r) = 2r = 4$, so $r = 2$: the third factor is $(x - 2)$.

Question 6 POL-07 · Polynomials, factor and remainder theorems · spec MM1.6

Find the quotient when $2x^3 - 3x^2 + 4x - 5$ is divided by $(x - 1)$.

Answer: C

$2x^3 - 3x^2 + 4x - 5 = (x - 1)(2x^2 - x + 3) - 2$: check the expansion

$2x^3 - x^2 + 3x - 2x^2 + x - 3 = 2x^3 - 3x^2 + 4x - 3$, then subtract 2 for the remainder.

Question 7 POL-11 · Polynomials, factor and remainder theorems · spec MM1.6

Find the coefficient of x^2 in the expansion of $(x + 1)(x - 2)(2x + 3)$.

Answer: A

$(x + 1)(x - 2) = x^2 - x - 2$; multiply by $(2x + 3)$: the x^2 term is $3x^2 - 2x^2 = x^2$, coefficient 1.

Question 8 POL-13 · Polynomials, factor and remainder theorems · spec MM1.6

Which of the following is a factor of $x^4 - 5x^2 + 4$?

Answer: E

$$x^4 - 5x^2 + 4 = (x^2 - 1)(x^2 - 4) = (x - 1)(x + 1)(x - 2)(x + 2).$$

Question 9 POL-14 · Polynomials, factor and remainder theorems · spec MM1.6

$p(x)$ is a cubic polynomial with $p(1) = p(2) = p(3) = 0$ and $p(0) = 12$. Find $p(4)$.

Answer: E

$$p(x) = a(x - 1)(x - 2)(x - 3); p(0) = -6a = 12 \Rightarrow a = -2. p(4) = -2 \cdot 3 \cdot 2 \cdot 1 = -12.$$

Question 10 POL-15 · Polynomials, factor and remainder theorems · spec MM1.6

The polynomial $x^3 + ax^2 + bx + c$ has roots 2, -1 and 3. Find the value of $a + b + c$.

Answer: D

$$p(1) = 1 + a + b + c, \text{ and } p(1) = (1 - 2)(1 + 1)(1 - 3) = (-1)(2)(-2) = 4. \text{ So } a + b + c = 3. \text{ (Directly: } (x - 2)(x + 1)(x - 3) = x^3 - 4x^2 + x + 6.)$$

Question 11 X2-08 · Polynomials, factor and remainder theorems · spec MM1.6

$p(x)$ is a polynomial. P : ' $p(2) = 0$ '. Q : ' $(x - 2)^2$ is a factor of $p(x)$ '. Which one of the following is true?

Answer: B

If $(x - 2)^2$ divides p then certainly $p(2) = 0$, so P is necessary for Q . But $p(x) = x - 2$ has $p(2) = 0$ without a repeated factor, so P is not sufficient.

Question 12 POL-06 · Polynomials, factor and remainder theorems · spec MM1.6

When $p(x)$ is divided by $(x - 1)$ the remainder is 3, and when divided by $(x + 1)$ the remainder is -1 . Find the remainder when $p(x)$ is divided by $(x^2 - 1)$.

Answer: B

The remainder has degree < 2 : $ax + b$ with $a + b = 3$ and $-a + b = -1$. So $b = 1$, $a = 2$: remainder $2x + 1$.